

Shagun Dwivedi

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EDUCATION

Indian Institute of Technology Madras, Chennai

2021 - 2025

Bachelor of Science, Data Science and Applications

CGPA: 9.04/10

Relevant Coursework: LLM, Deep Learning for Computer Vision, Reinforcement Learning, Linear Statistical Models, Machine Learning Techniques, Mathematics for Data Science I & II, Statistics for Data Science I & II

University of Lucknow, Lucknow

2020 - 2023

Bachelor of Arts (Hons.) Ancient Indian History and Archaeology

CGPA: 8.04/10

PUBLICATIONS

- **Dwivedi, S.**; Saquib, M.; Gopalan, K. “Performance of Grapheme-Based Tokenizers on Word-level and Natural Language Understanding tasks for Indic Languages”. *Findings of the Association for Computational Linguistics: EMNLP 2026 (In Press)*
- **Dwivedi, S.**; Gopalan, K. “Comparative Analysis of the Intrinsic Metrics for Tokenizers and their Effect on Downstream Tasks for Hindi and Marathi”. *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*
- **Dwivedi, S.**; Gopalan, K. “Synthetic Data Generation using Grapheme-level Visual Units for Handwritten Text Recognition of Grantha Manuscripts”. *Second Workshop on Curated Data for Efficient Learning (CDEL), ECCV 2026*
- Chincholikar, K.*; **Dwivedi, S.***; Gopalan, K.; Awasthi, T. “A Case Study of Handwritten Text Recognition from Early Modern Sanskrit Manuscripts”. *2025 Proceedings of the Computational Sanskrit & Digital Humanities at the 19th World Sanskrit Conference.*
- Jain, R.; **Dwivedi, S.**; Gopalan, K. “Converting Gujarati Text in Custom-Embedded Subsetted Non-Unicode Fonts to Searchable Formats: A Case Study Using Jain Religious Texts”. *Computer Vision and Image Processing 2024. CCIS vol. 2475, Springer.*
- Vaishampayan, J.; Gopalan, K.; **Dwivedi, S.** “Synthetic Line Image Generation from Cropped Grapheme Images for Handwritten Marathi Text Recognition”. *ICDAR 2026 Workshop on Documents Analysis of Low-resource Languages (In Press)*
- Valaboju, B.; **Dwivedi, S.**; Chincholikar, K.; Gopalan, K.; Vidwans, V. “A Semi-Automatic Text Recognition Tool for Pre-Colonial Handwritten Manuscripts in Devanāgarī Script”. *HCI International 2025 Posters. CCIS vol. 2529, Springer.*
- **Dwivedi, S.**; Deshpande A.; Gopalan, K.; Bhuskute, A. “An Adaptive AI Tool for Introductory French and German Courses”. *Proceedings of 2025 6th International Conference on Artificial Intelligence in Education Technology.*

RESEARCH EXPERIENCE

AI Engineer

March 2024 - Present

Centre for Interdisciplinary AI, FLAME University

- Conducting research across NLP and document understanding, for low-resource Indic languages, exploring how script-specific properties inform model design across modalities.
- Proposed and evaluated grapheme-aware tokenization schemes for 11 Indic languages, analyzing intrinsic tokenizer metrics and their downstream effects on model performance; work published at ACL 2026 (Main), and EMNLP 2026 (Findings).
- Built Handwritten Text Recognition and text extraction pipelines for low-resource and historical Indic scripts grounded in script-specific properties; work published at ECCV 2026 (workshop), ICDAR 2026 (workshop), WSC 2025, HCII 2025, and CVIP 2024.

PROJECTS

Grapheme based Tokenization for Indic languages 🌐

- Proposed a novel grapheme-aware tokenization scheme for Hindi and Marathi, in Devanagari script, achieving competitive performance to popular tokenizers on intrinsic & extrinsic metrics; work accepted to ACL 2026 (Main).
- Pretrained monolingual and multilingual T5 models using different tokenizers for question answering, grapheme-to-phoneme conversion, and transliteration tasks.
- Extended the published work based on the tokenization of Devanagari script to 9 other Indic languages, demonstrating competitive performance on word-level and NLU based tasks. Paper based on this work accepted to EMNLP 2026 (Findings).
- Developed a lexical, semantic, and syntactic similarity based automated question answering evaluation framework by training FastText word2vec embeddings and using stanza for POS-tagging.

Synthetic Data Generation for Handwritten Text Recognition of Low-resource Indic Scripts

- Built an automated handwritten glyph extraction and line image generation to close the domain gap between synthetic and real manuscript images.
- Developed a synthetic data generation pipeline using grapheme-level visual units to enable HTR for Grantha manuscripts with as few as 5 annotated pages, addressing the challenge of large-scale manual annotation.
- Trained a CRNN-CTC based recognition model on synthetically generated line images, achieving competitive performance despite extremely limited real training data.

Converting Custom-Embedded Subsetted Non-Unicode Fonts to Searchable Formats

- Devised text reconstruction method to handle inconsistent glyph-mapping and non-canonical Unicode sequences between legacy font embeddings and Unicode representation.
- Worked on glyph feature extraction using VGG16, glyph-Unicode mapping, and glyph annotation for Hindi, Marathi, Bangla, and Gujarati fonts.
- Proposed method performs better than state-of-the-art systems, achieving character error rate of 0.079% to 2.02% and grapheme cluster error rate of 0.095% to 2.85% across documents with varied layouts.

Handwritten Text Recognition for Pre-Colonial Sanskrit manuscripts

- Finetuned a ResNet-BiLSTM-CTC based recognition model outperforming Document AI and Tesseract on character error rate (58%-90% improvement) and grapheme cluster error rate (42-79% improvement) across three manuscripts.
- Adapted grapheme cluster error rate for Devanagari to provide stricter evaluation of conjunct and diacritic recognition errors at visual character boundaries.
- Built components of a Vue.js and Flask based application to digitize, manually annotate and transliterate Sanskrit manuscripts.

ACHIEVEMENTS

- Secured an All India Rank of 488 (top 1.25%) among 39,210 candidates in GATE Data Science (DA), 2024.
- Winner of “Best New Video Innovation” and \$5,000 prize at Adobe Express Add-ons Hackathon 2025. [🔗](#)
- Awarded Certificate of Merit by IIT Madras, for maintaining CGPA above 9.0, placed in top 5% of the graduating cohort.
- First Runner up in Hack4Change Hackathon conducted during IITM Paradox in Margazhi 2024.
- Awarded the Chhatra Kalyan Scholarship (University of Lucknow) and the VPROPEL Scholarship (Verizon India in collaboration with IIT Madras).
- Invited to participate in Sakura Science Exchange Program hosted by JST, Japan (2019).

SKILLS

Languages: Python, R, Java, SQL, JavaScript

AI & ML: PyTorch, TensorFlow, HuggingFace Transformers, FastText, OpenCV, pandas, numpy, scipy, scikit-learn

Development: Flask, FastAPI, Vue.js, React

Tools & Platforms: Git, Docker, AWS, GCP, Azure, Weaviate, Firestore

POSITIONS OF RESPONSIBILITY

Teaching Assistant

Dec. 2022 - Dec. 2023

Modern Application Development I & Modern Application Development II, IIT Madras

- Assessed 200+ student projects in Vue.js, Flask, and Redis and prepared evaluation rubrics over three course terms.

Head – Editing & Documentation Team

Jul. 2023 - Jul. 2024

Sahityika, Literary Society of IIT Madras BS Degree

- Handled editorial work related to event organisation, outreach and creative publications for the literary society.